

LadyBug Tracker



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Introduction

LadyBug Tracker is a plugin that integrates bug tracker systems with Unreal Engine. This plugin makes your work much more efficient and convenient. You do not have to switch all the time between the Unreal Engine editor and a bug tracker page in the web browser. With LadyBug Tracker, you can do these tasks:

- View and edit the details of reported issues easily.
- Browse the issues directly in the level viewport.
- Report a bug directly from the editor and from the game.
- Make Blueprints that report bugs automatically for you.

Data Storage and Privacy

The LadyBug Tracker plugin does not store any data. The plugin sends all reports directly to the databases of the integrated services: Jira, YouTrack, Mantis, Trello, Asana, HacknPlan, Redmine, ClickUp and Notion. It uses the API of each service. Only you have access to this data. The plugin never sends it to a different place. Also, the plugin does not collect data for statistics or diagnostics.

Features list

- Browse the issues with a fast and powerful filter.
- 3D markers. View the issue markers on a level.
- "Go to" button. One click opens the level at the location of the issue.
- Edit the details of issues.
- Report issues in the game.
- Report issues in the editor.
- Report crashes.
- Capture screenshots and logs.
- Capture the level and the camera position.
- Fill the fields automatically.
- Edit screenshots.
- Pending issues list. Complete the details of new issues before you submit them.
- View the issue details (general fields, custom fields, notes, attachments, history).
- Add notes.
- Add attachments.

Supported Integrations

LadyBug Tracker is designed to support many bug tracker systems.

YouTrack

- **Integration Documentation**
- <https://www.jetbrains.com/youtrack/>
- Free for up to 10 users
- Project management tool
- Strongly recommended

Mantis Bug Tracker

- **Integration Documentation**
- <https://www.mantisbt.org/>
- Open source. Free to use
- Designed for bug tracking

MantisHub

- **Integration Documentation**
- <https://www.mantishub.com/>
- A hosted version of Mantis Bug Tracker
- Designed for bug tracking

Trello

- [Integration Documentation](#)
- <https://trello.com/>
- Free for up to 15 users
- Very popular in game development

Jira Software

- [Integration Documentation](#)
- <https://www.atlassian.com/software/jira>
- Free for up to 10 users
- Project management tool

Jira Server

- This integration is in the beta stage. It needs Jira Server 10.3.7 or higher
- The integration does not support labels
- A paid version that you host yourself
- A project management tool that is popular in big companies

Asana

- [Integration Documentation](#)
- <https://asana.com>
- Free for up to 10 users
- Project management tool

HacknPlan

- [Integration Documentation](#)
- <https://hacknplan.com/>
- Unlimited users

Redmine

- [Integration Documentation](#)
- <https://www.redmine.org/>
- Open source. Free to use
- Project management web application.

ClickUp

- [Integration Documentation](#)
- <https://clickup.com/>
- A free plan is available
- Project management tool. On a free plan, the plugin keeps the Level and Camera values in the task description, not in custom fields

Notion

- [Integration Documentation](#)
- <https://www.notion.so/>
- A free plan is available
- Workspace and documentation tool.

Do you need a different integration? Tell me.

Integrations comparison

	YouTrack	Trello	Jira Software	Jira Server	Mantis	MantisHub	Asana	HacknPlan	Redmine	ClickUp	Notion
View issues	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Edit issues	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Report issues	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Custom fields	Yes	Yes ¹	Yes	Yes	Yes	Yes	Yes ¹	No ²	Yes	Yes ¹	Yes
Sort by columns	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Filter issues	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
3D Markers	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
View attachments	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Add attachments	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Delete attachments	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No ²	Yes
View comments	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Add comment	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Delete comment	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No ²
View checklists (subtasks)	No ²	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes
Edit checklists (subtasks)	No ²	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes
Create project	No ²	Yes	No	No	Yes	Yes	Yes	Yes	No	Yes	No ²
View history	No	No	No	No	Yes	Yes	No	No	No	No	No
Free version users	10	15	10	-	unlimited	0	10	unlimited	unlimited	unlimited	unlimited ⁴
Free version storage, total / per file	30 GB / 500 MB	unlimited / 10 MB	2 GB / 2GB	-	unlimited / unlimited	0 / 0	unlimited / 25MB	500 MB / 5 MB	unlimited / unlimited	1 GB / 100 MB	unlimited / 5 MB
Free version projects	unlimited	10	unlimited	-	unlimited	0	unlimited	unlimited	unlimited	unlimited lists (max 5 Spaces)	unlimited

¹ Available in the paid version of the service

² The service does not have this feature

³ Not implemented yet

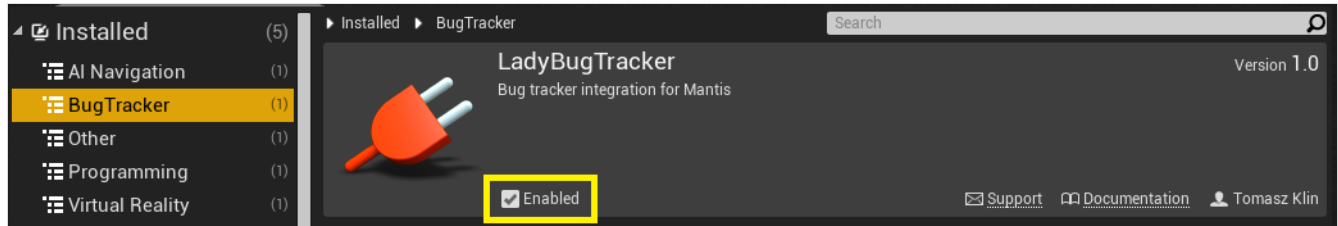
⁴ When a workspace has more than one member, Notion sets a limit of 1,000 blocks

Getting Started

Enabling Plugin

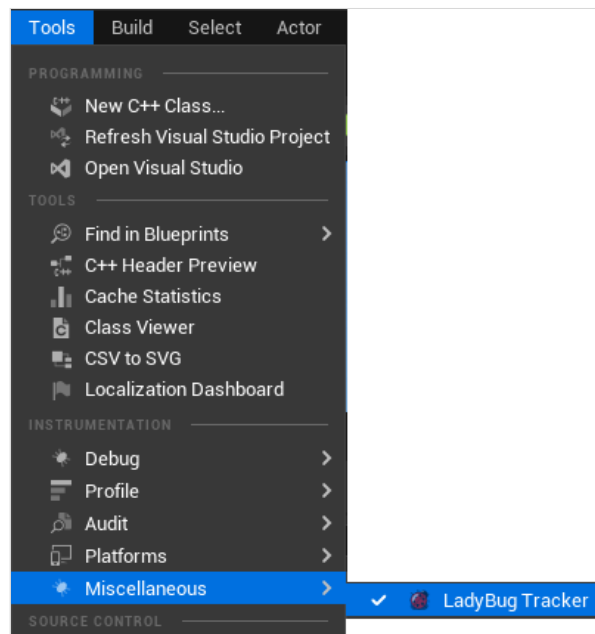
To enable the plugin, do these steps:

1. Go to Edit -> Plugins.
2. Select the **Bug Tracker** category.
3. For LadyBug Tracker, select the **Enabled** check box.



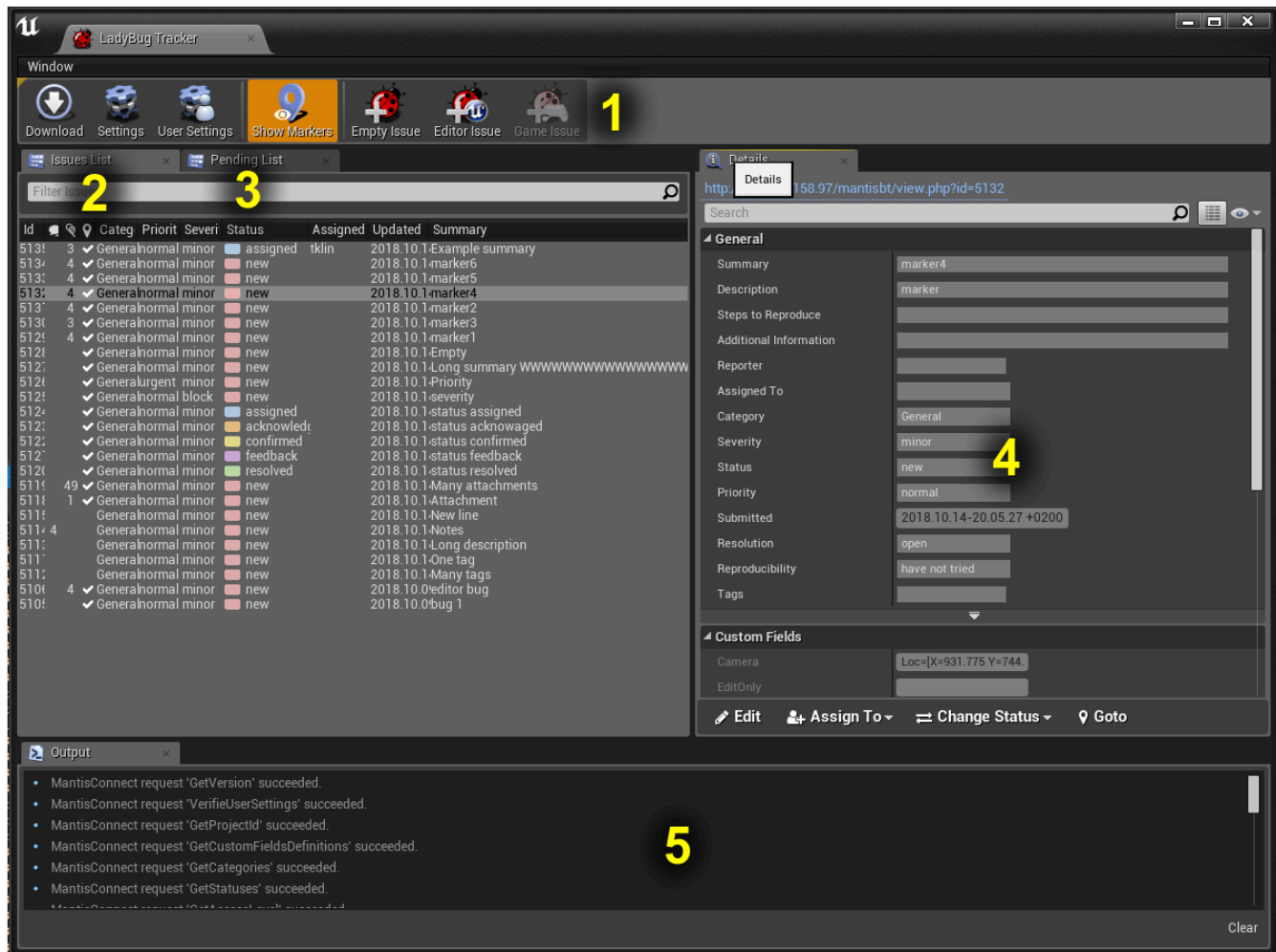
Opening Tool Window

LadyBug Tracker is a tab in Unreal Editor. To open the tab, use the Window menu.



Interface

The LadyBug Tracker user interface has these parts:



1. Toolbar
2. Issues List
3. Pending List
4. Details
5. Output

Toolbar



1. Download - Downloads all the issues from the server.
2. Settings - Opens the [Project Settings](#) window.
3. User Settings - Opens the [login window](#).
4. Show Markers - Shows or hides the markers in the Level Viewport (see [3D Markers](#)).

Issues List

Pending List

Filter Issues

Path	Priority			Categor	Severity	Handler	Summary
../Saved/Bugs/Pending/2018.10.09-20.01.04	normal			General	minor		
../Saved/Bugs/Pending/2018.10.09-20.49.03	normal			General	minor		
../Saved/Bugs/Pending/2018.10.09-20.49.04	normal	1		General	minor		
../Saved/Bugs/Pending/2018.10.09-20.49.07	normal	49		General	minor		
../Saved/Bugs/Pending/2018.10.09-20.49.08	normal	49		General	minor		
../Saved/Bugs/Pending/2018.10.09-20.49.09	normal	49		General	minor		
../Saved/Bugs/Pending/2018.10.09-20.49.16	normal			General	minor		
../Saved/Bugs/Pending/2018.10.09-20.49.34	normal			General	minor		
../Saved/Bugs/Pending/2018.10.09-20.49.37	normal	1		General	minor		
../Saved/Bugs/Pending/2018.10.12-20.41.24	normal			General	minor		
../Saved/Bugs/Pending/2018.10.14-20.03.04	normal			General	minor		marker

Details Panel

The Details panel shows all the available information about the selected bug.

Details

<http://104.238.158.97/mantisbt/view.php?id=5135>

Search

General

Summary

Example summary

Description

Some description

Steps to Reproduce

Additional Information

Reporter

Assigned To

tklin

Category

General

Severity

minor

Status

assigned

Priority

normal

Submitted

2018.10.14-2018.10.14 14:11 +0200

Resolution

open

Reproducibility

always

Tags

Custom Fields

Attachments

test

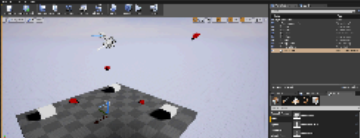
2018.10.14-18.14.13

Log00000.log (69030 bytes) text/plain

test

2018.10.14-18.14.20

ScreenShot00000.png (1476538 bytes) image/png



Edit
Assign To
Change Status
Goto

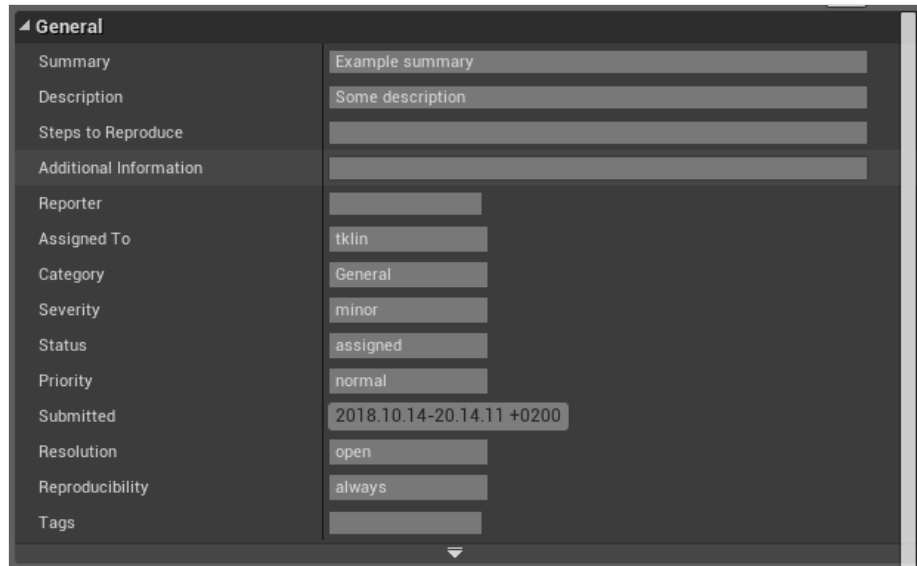
1. A link that opens the selected issue in the web browser.
2. Property list (see [Property List](#)).
3. Toolbar (see [Details Panel Toolbar](#)).

Property List

The property list contains all the most important information about the selected issue.

General

Standard properties are available by default in Mantis.

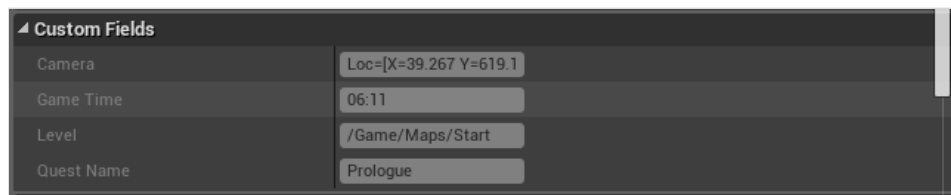


A screenshot of the 'General' property list panel in Mantis. The panel has a dark grey header with a left-pointing triangle and the word 'General'. Below the header is a list of properties, each with a label on the left and a text input field on the right. The properties and their values are: Summary (Example summary), Description (Some description), Steps to Reproduce (empty), Additional Information (empty), Reporter (empty), Assigned To (tklin), Category (General), Severity (minor), Status (assigned), Priority (normal), Submitted (2018.10.14-20.14.11 +0200), Resolution (open), Reproducibility (always), and Tags (empty). The panel has a vertical scrollbar on the right side.

Property	Value
Summary	Example summary
Description	Some description
Steps to Reproduce	
Additional Information	
Reporter	
Assigned To	tklin
Category	General
Severity	minor
Status	assigned
Priority	normal
Submitted	2018.10.14-20.14.11 +0200
Resolution	open
Reproducibility	always
Tags	

Custom Fields

Fields that the user defines. The plugin shows custom fields only as text. The server checks the values when you submit the issue.

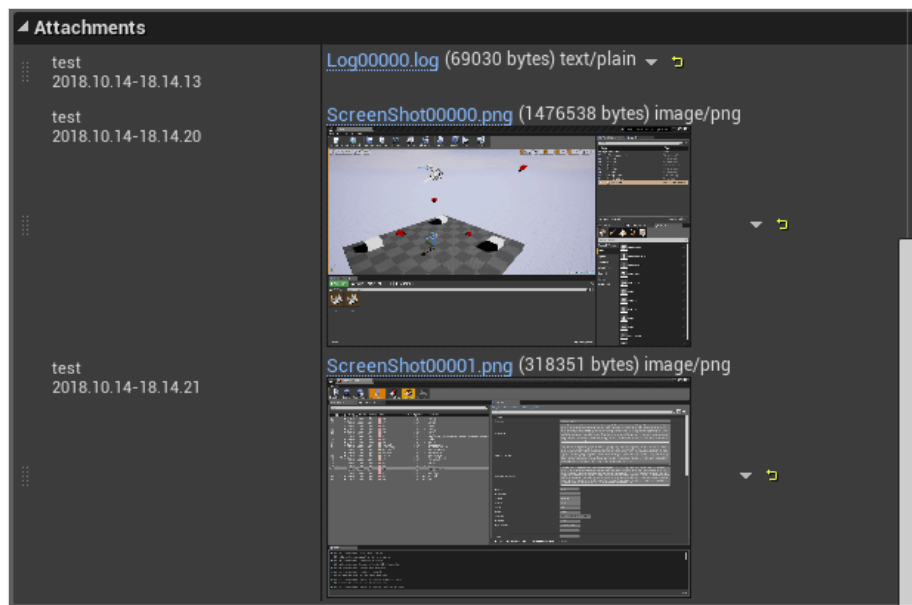


A screenshot of the 'Custom Fields' panel in Mantis. The panel has a dark grey header with a left-pointing triangle and the words 'Custom Fields'. Below the header is a list of custom fields, each with a label on the left and a text input field on the right. The fields and their values are: Camera (Loc=[X=39.267 Y=619.1]), Game Time (06:11), Level (/Game/Maps/Start), and Quest Name (Prologue). The panel has a vertical scrollbar on the right side.

Field	Value
Camera	Loc=[X=39.267 Y=619.1]
Game Time	06:11
Level	/Game/Maps/Start
Quest Name	Prologue

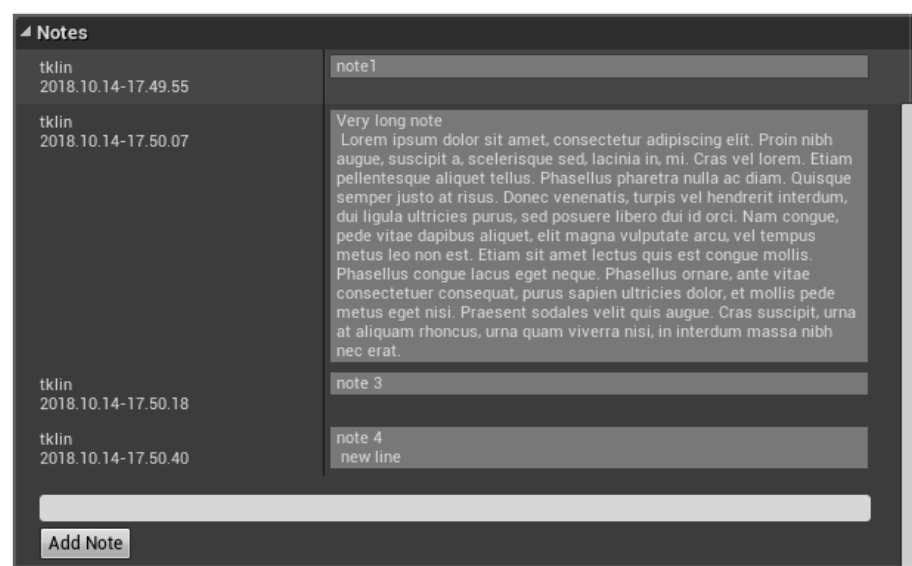
Attachments

The Attachments section shows a list of attachments. For PNG, BMP and JPG images, the plugin can show thumbnails.



Comments

The Comments section shows all the notes of the issue. You can also add new notes here.



History (only mantis integration)

The history of all the activities on the issue.

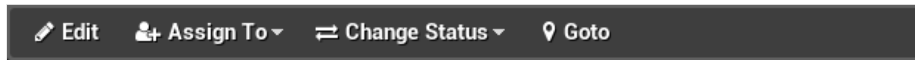
Note: At this time, the history does not show readable values for enum types, for example category or priority. It shows the numeric value.

History			
Date Modified	User Name	Field	Change
2018.10.14 18:05	test	New Issue	
2018.10.14 18:05	test	File Added	Log00000.log
2018.10.14 18:05	test	File Added	ScreenShot00000.png
2018.10.14 18:05	test	File Added	ScreenShot00001.png
2018.10.14 18:05	test	File Added	ScreenShot00002.png

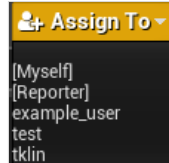
Details Panel Toolbar

The toolbar shows a different set of controls in each mode.

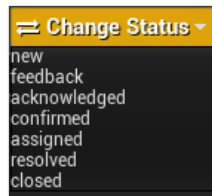
View Mode



1. Edit - Starts the edit mode.
2. Assign To - Assigns the issue to a user quickly.



1. Change Status - Changes the status of the issue quickly.



1. Goto - Opens the level at the location that the issue specifies (see [3D Markers](#)).

Edit Mode



1. Cancel - Stops the edit mode. The plugin discards all the changes.
2. Update - Submits all the changes to the server.

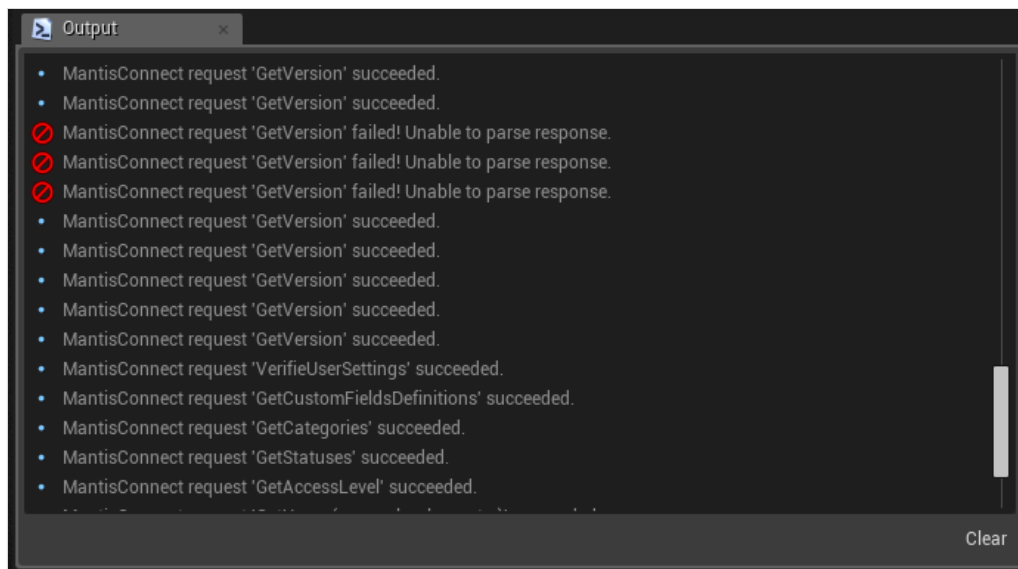
Report Mode



1. Discard - Deletes the issue from the pending list.
2. Submit Issue - Submits the issue to the server.

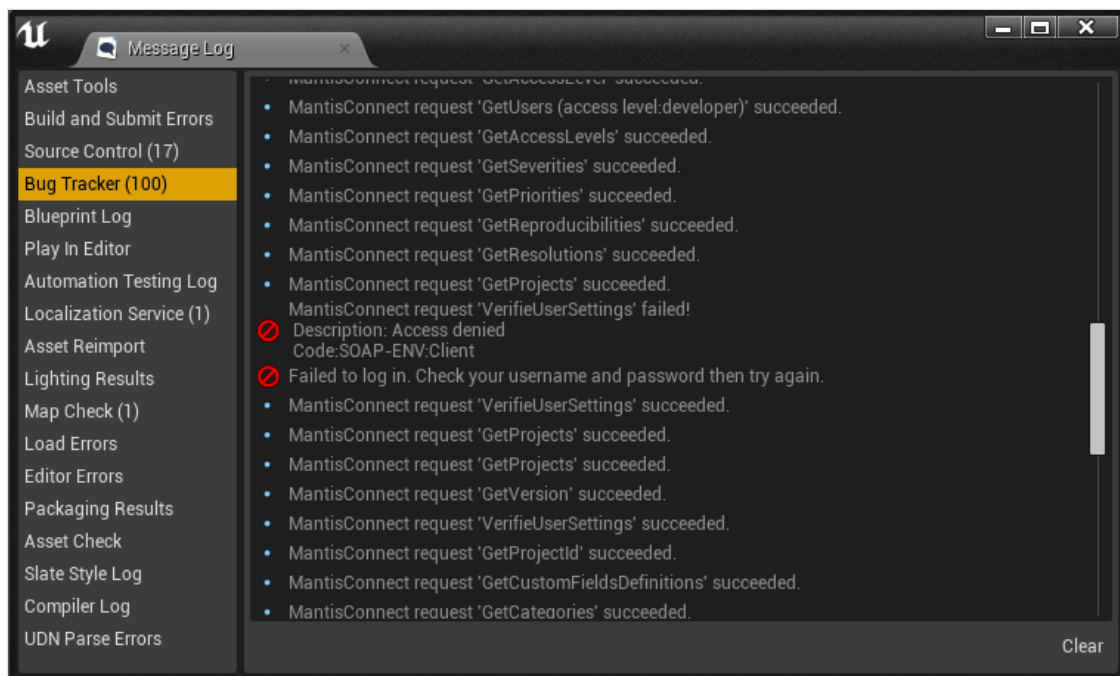
Output

The Output panel collects the log output of all the plugin operations. If a problem occurs, look for the cause here first.

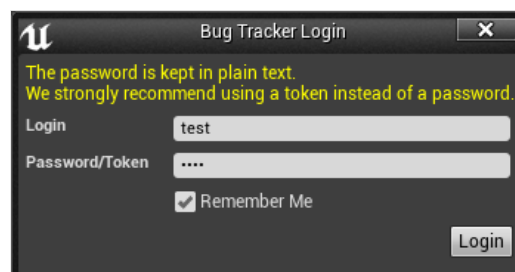


Message Log

The Message Log tab also shows the logs from the [Output panel](#), in the Bug Tracker section. To make the logs more detailed, use the “Debug” section of the project settings.



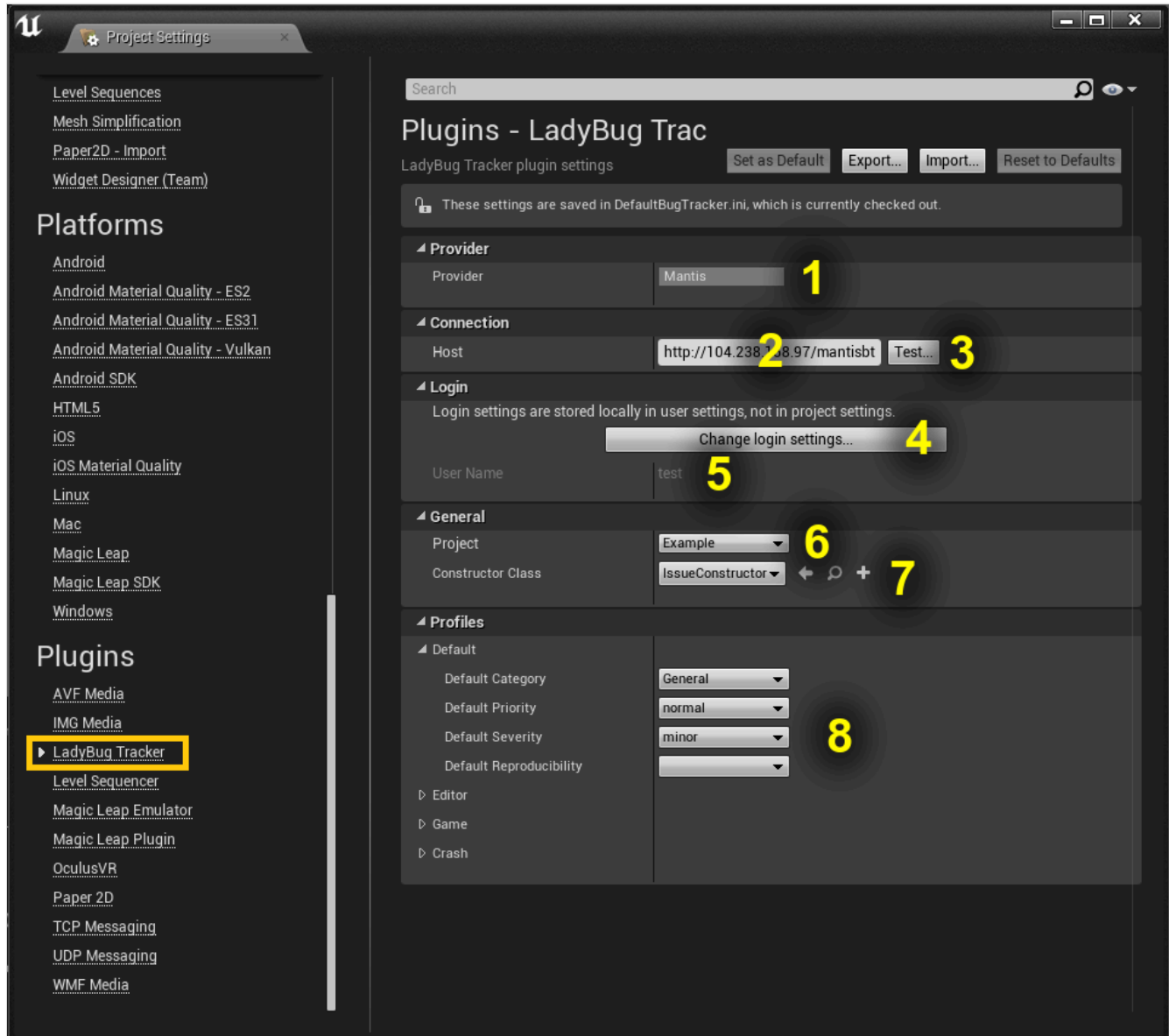
Login Dialog Box



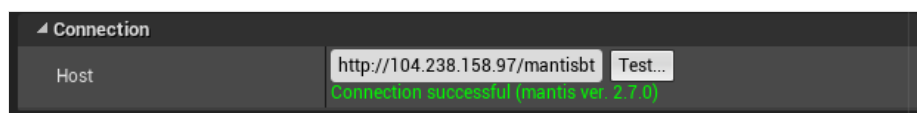
This window can be different for each provider that you select.

Project Settings

You can edit the LadyBug Tracker settings in the Project Settings tab, in the Plugins section. The settings are in the **/Config/DefaultBugTracker.ini** file. Your team can share this file.



1. Provider - The bug tracker provider.
2. Host - The address of the bug tracker host.
3. Test - Tests the connection to the host.



1. Change Login Settings - Opens the login window (see [Login Dialog Box](#)).
2. User Name - The user that is logged in now.
3. Project - Lets you select a project that is available to the logged-in user.

4. Constructor Class - The issue constructor class (see [Issue Constructor](#)).
5. Profiles - The default values for each issue profile (see [Profiles](#)).

User Settings

The plugin keeps the user settings in the **/Saved/Config/WindowsEditor/BugTrackerPerProjectUserSettings.ini** file. The user settings contain, for example, the preferences, the login and the password.

Markers

When you report a bug in the editor or in the game, the plugin automatically saves the level name. It also saves the location and orientation of the camera. The Issues List marks each bug that has this information, in the column with this symbol:



Attention! This feature is available only if you add the Camera and Level fields to the custom fields.

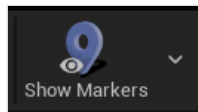
Goto

An issue keeps location information. With this information, you can go very quickly to the level, at the location of the issue. To do this, click this button on the [Details Panel Toolbar](#):



3D Markers

The level viewport can show the markers of all the issues for the level that is open now. To enable this feature, select this button in the [Toolbar](#):



If you select a marker in the level viewport, the Details panel opens its issue. If you double-click a marker, the viewport goes to the location of the marker.



You can change the labels of the 3D markers here:



Show Markers

▼

▼ 3D Marker

Show Label	☒	
Color Scheme	Priority	▼
▶ Label Color		
Max Label Len...	50	

Reporting Bugs in Editor

The most powerful feature of this plugin is how you report issues. It is very convenient, and you can customize it.

The plugin sends all reported bugs to the pending list. Thus you can describe an issue and submit it to the server at any time. For each new bug, the plugin makes a subdirectory in the `../Saved/Bugs/Pending/` directory. The plugin automatically adds all the files in this subdirectory to the bug as attachments.

Reporting Editor Issue

You can report an issue in two ways: from the [Toolbar](#), or with a keyboard shortcut. To use a keyboard shortcut, first assign the command to a key. The bugs that you report in the editor automatically contain the log and the screenshots of all the open editor windows.

Reporting Game Issue

You can report a bug in the game in three ways: from the [Toolbar](#), from the console, or from a Blueprint (see [Triggering Reporting Bugs](#)). To report a bug from the console, do these steps:

1. Open the console in the game.
2. Enter the command **ReportBug bug_summary**.

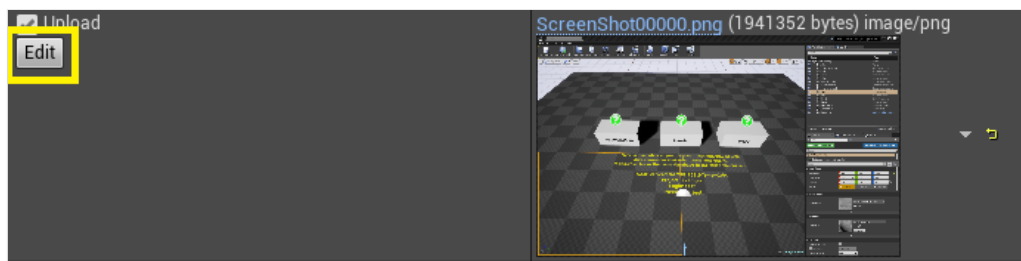
The bugs that you report from the game have a screenshot and the log as attachments.

Reporting Crashes

Each crash also appears on the pending list. Thus it is very easy to report a crash. A bug from a crash contains the log from the game and the minidump. It also contains the other files that the crash reporting tool of Unreal Engine saved.

Editing Screenshots

Before you submit an issue, you can edit all the screenshots in an image editor that you set, for example Paint.



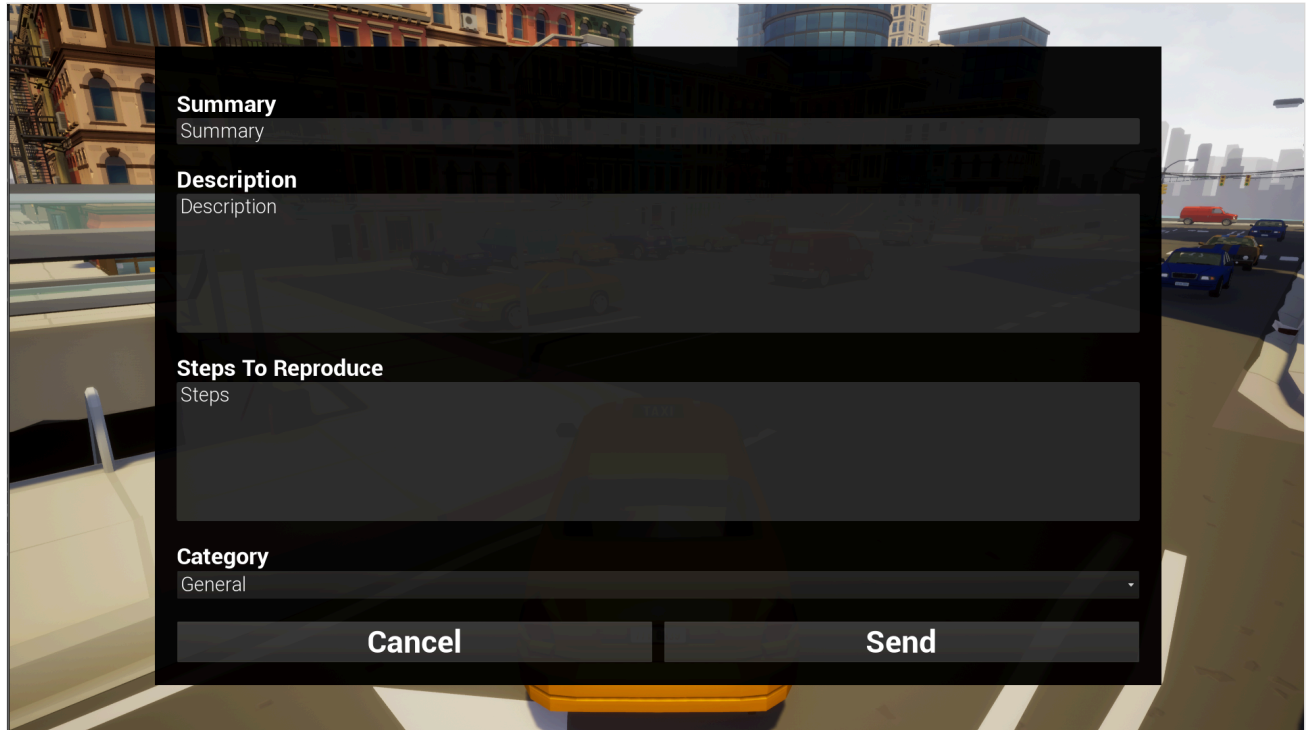
Use the **Upload** check box to select the files to attach.

Reporting User Feedback

LadyBug Tracker lets the shipping game send feedback from users, with attachments (logs, saves, screenshots).

User feedback uses different authentication data from the editor. For bug reports from users, it is good practice to use a separate account that you make only for this purpose. For security, give this account the minimum permissions.

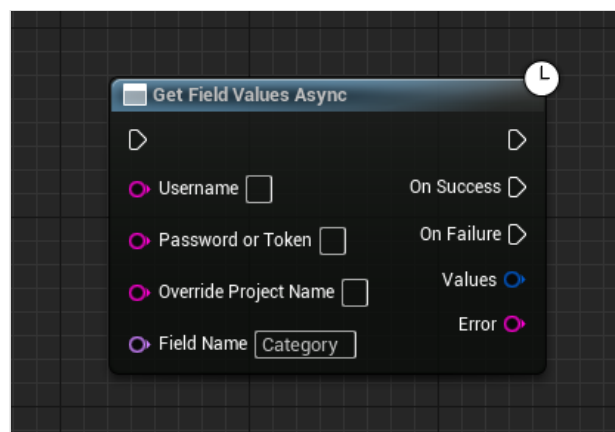
1. Make a feedback form. It can look like the form that comes with the plugin (see \LadyBugTracker\Content\UIExample\ReportForm).



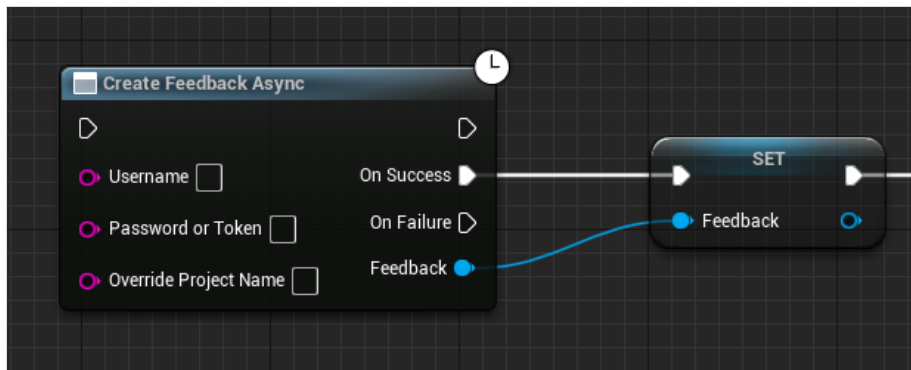
The image shows a feedback form titled "ReportForm" overlaid on a game scene. The form has the following sections:

- Summary**: A text input field with the placeholder text "Summary".
- Description**: A large text input field with the placeholder text "Description".
- Steps To Reproduce**: A large text input field with the placeholder text "Steps".
- Category**: A dropdown menu with "General" selected.
- Buttons**: "Cancel" and "Send" buttons at the bottom.

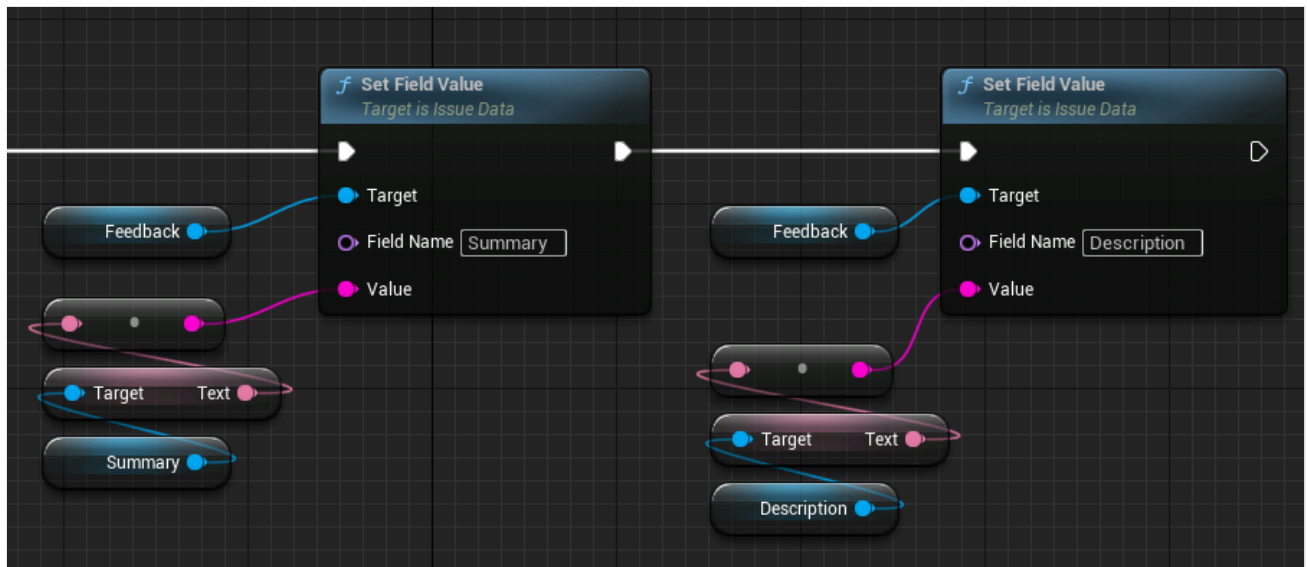
2. To fill your UI with the possible values of a field, use the **Get Field Values Async** node.



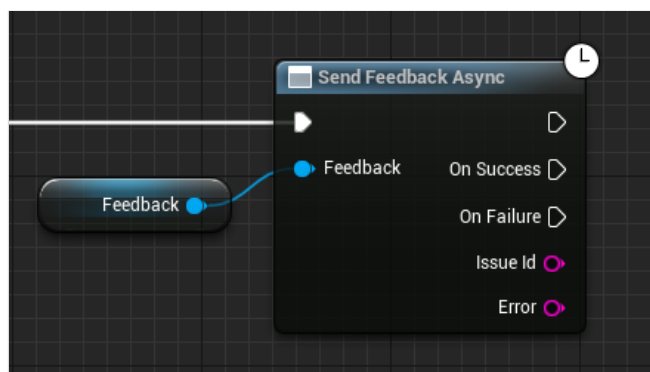
3. Make a feedback object with **Create Feedback Async**.



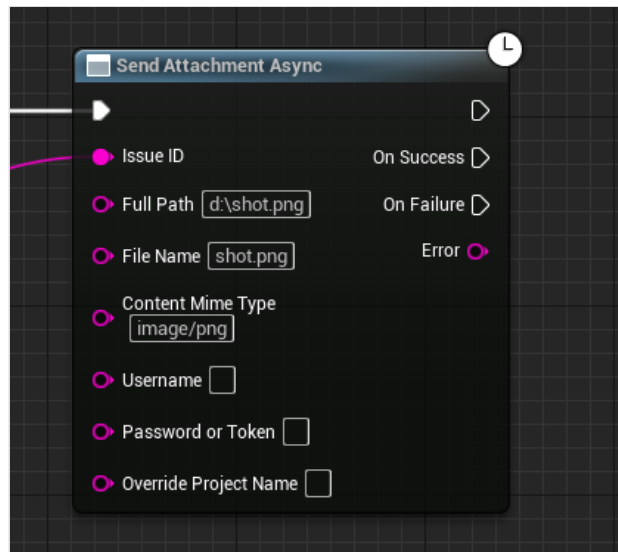
4. Set all the data that you want to give.



5. Send the feedback to the server with **Send Feedback Async**.

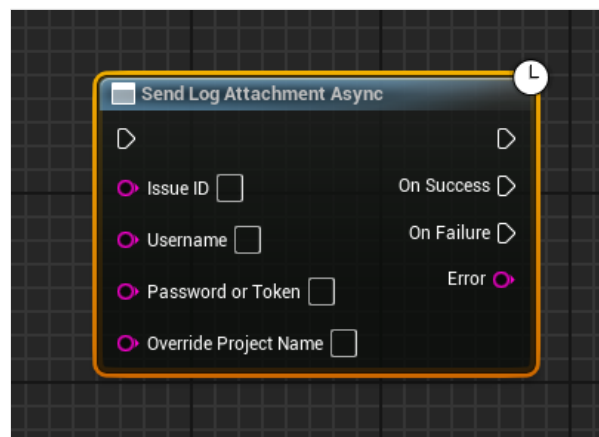


6. Send Feedback returns an Issue Id. Use this Issue Id to upload attachments to the issue with **Send Attachment Async**.



7. The game can already have data in memory, for example a save, a generated report or a screenshot that you took. To attach this data, use **Send Attachment From Memory Async** instead. This node takes the same Issue Id, file name and mime type. In place of the path, it takes a byte array.

8. To attach the game log, use **Send Log Attachment Async**. This node collects the log itself. Thus it needs only the Issue Id.



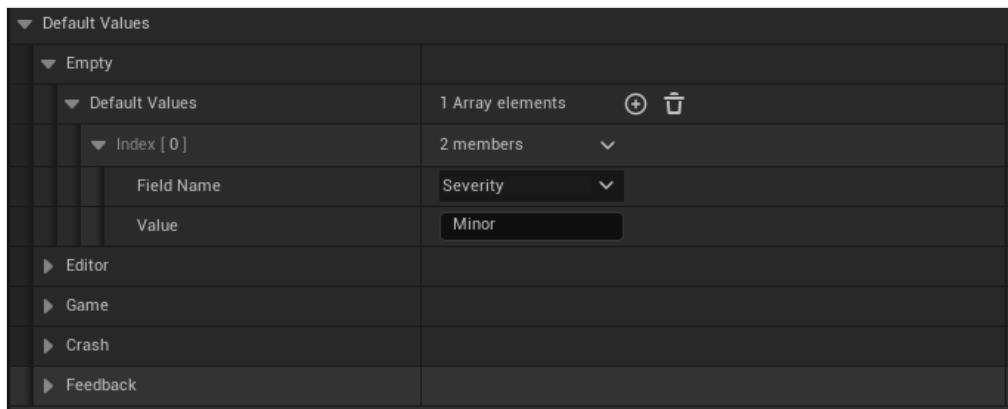
Each node in these steps also has a synchronous variant. [Blueprint API](#) shows both variants.

Reporting Automation

To make the QA team more efficient, the plugin can do some of the tedious work automatically.

Profiles

Profiles let you override the default values of fields for each report mode. To override these default values, open the LadyBug Tracker [Project Settings](#).

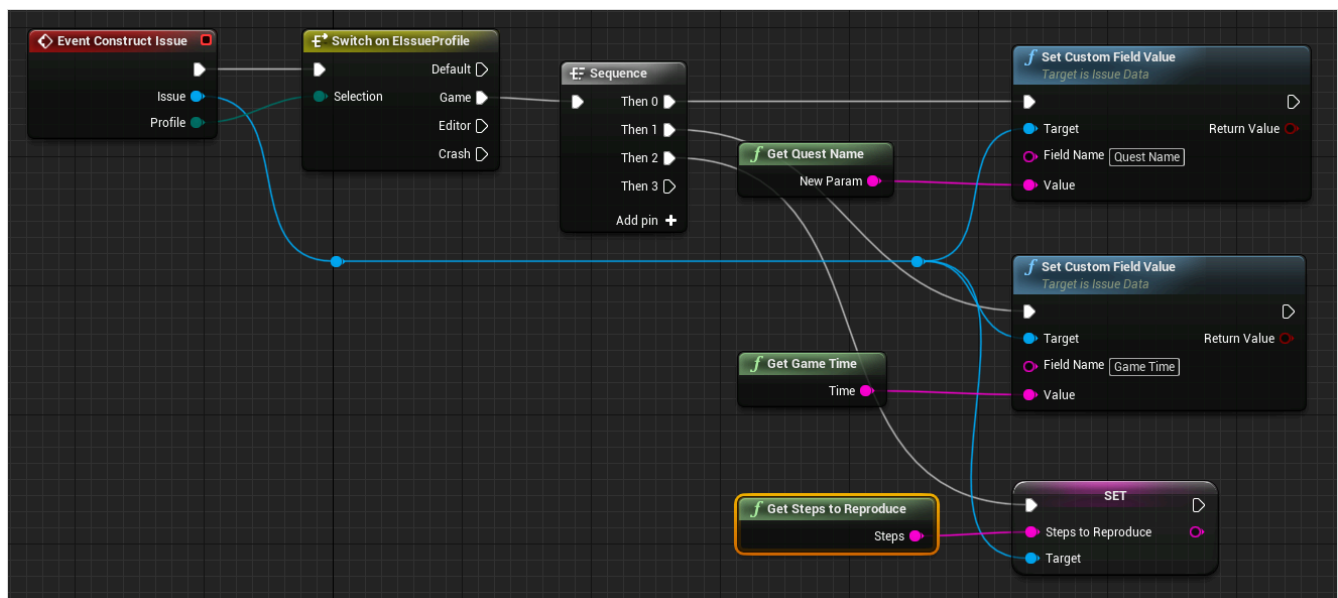


Issue Constructor

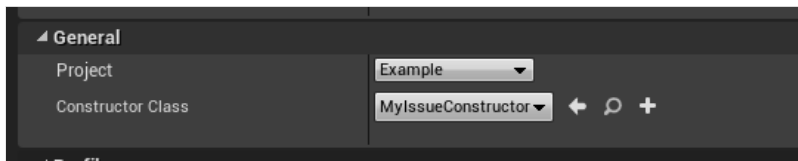
Issue Constructor is a Blueprint class that can change any parameter of an issue. For example, you can add information that helps to reproduce the issue. This can be the name of the current quest or the skills of the player character.

To make your own Issue Constructor, make a Blueprint that inherits from **IssueConstructor**. In this Blueprint, override the **Construct Issue** function. The plugin calls **Construct Issue** for all bugs, immediately after each report.

In the example below, the constructor adds information to all the issues that you report in the game. It adds the current level, the game time and some steps to reproduce.

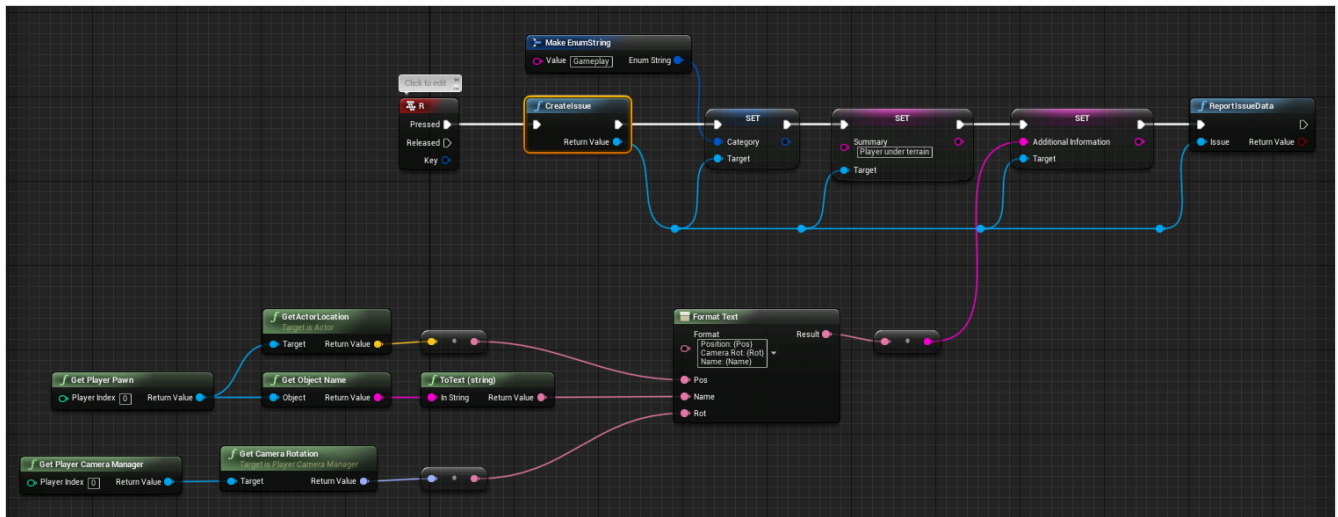


To set your Issue Constructor, select an IssueConstructor class in the combo box in [Project Settings](#).



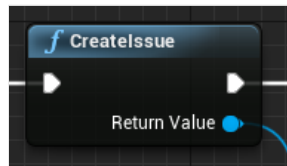
Triggering Reporting Bugs

You can also make issues automatically when a specified condition occurs in the game. For example, the game can make an issue when the player falls under the terrain.

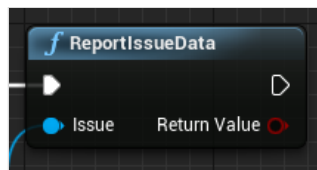


With this method, you can add more information to the report, for example the player position or the objects that collide.

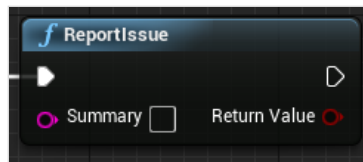
To report a bug, call the **Create Issue** node. This node only makes an Issue object. You can change this object later.



When the Issue object is ready, call the **Report Issue Data** node. The reported issue goes to the [Pending List](#).



You can also call **Report Issue**. This node makes and reports an issue with the summary that you give.



Blueprint API

All the nodes that the plugin adds are in the **Bug Tracker** category. Each node that communicates with the server has two variants:

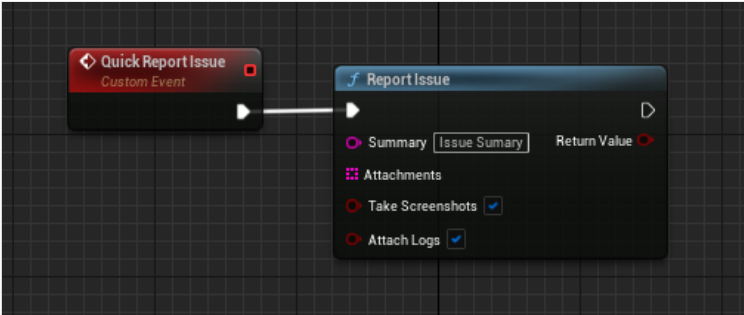
- **Synchronous** - The node returns immediately. It reports the result through its **On Success** and **On Fail** delegate pins. Bind these pins to your own custom events. Examples 1 to 4 below use this variant.
- **Asynchronous** - The node name ends with **Async**. In place of the delegates, the node has **On Success** and **On Failure** execution pins. These pins fire when the server answers. You do not have to bind anything manually. Thus these nodes are easier to start with. Example 5 shows the same graph with them.

Node	Asynchronous variant	What it does
Create Issue	-	Makes an empty issue. You complete it, then give it to Report Issue Data .
Report Issue Data	-	Sends an issue from Create Issue to the pending list.
Report Issue	-	Makes and reports an issue with a given summary in one step.
Create Feedback	Create Feedback Async	Makes the feedback object that your form completes.
Send Feedback	Send Feedback Async	Submits the feedback and returns the Issue Id.
Send Attachment	Send Attachment Async	Uploads a file from disk to that issue.
Send Attachment From Memory	Send Attachment From Memory Async	Uploads a byte array to that issue. It does not need a file on disk.
Send Log Attachment	Send Log Attachment Async	Uploads the current game log to that issue.
Get Field Values	Get Field Values Async	Returns the possible values of a field. Use them to fill your own UI.
Get Issues	Get Issues Async	Returns the issues of the project. The game can then show them in its own list.
Get Bug Tracker Name	-	Returns the name of the configured integration.

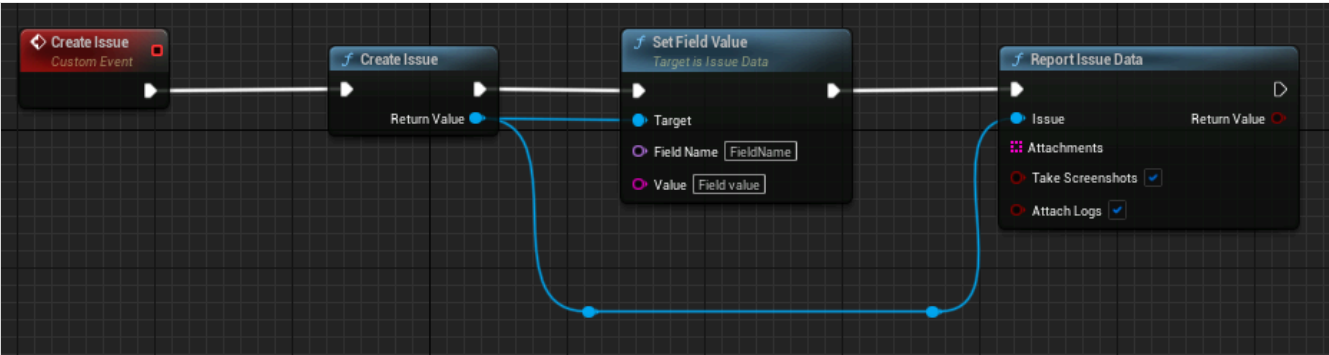
Some nodes ask for **Username** and **Password or Token**. For these, use the separate account from [Reporting User Feedback](#). Never use your own account. If **Override Project Name** is empty, the node uses the project from [Project Settings](#).

Blueprints example

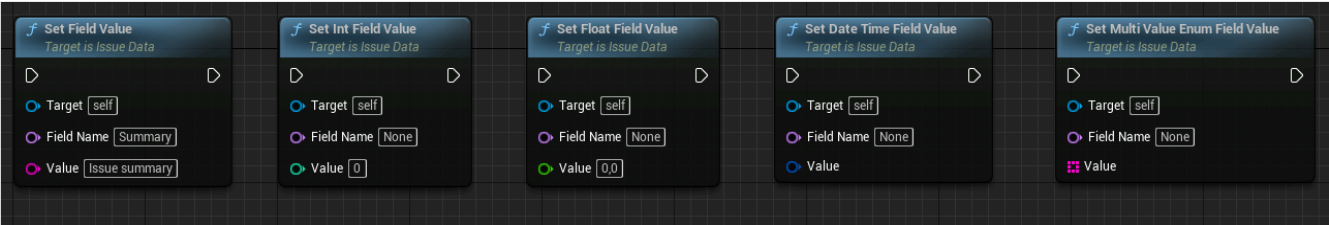
Example 1 - Quick report issue



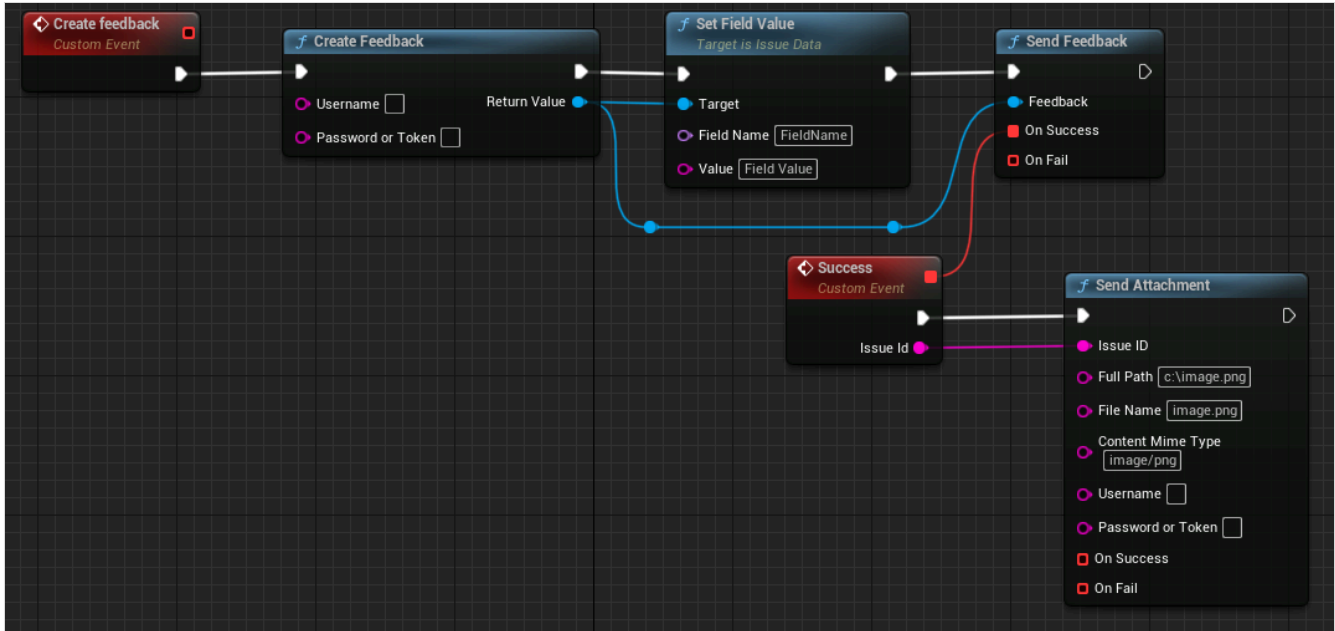
Example 2 - Report issue



Example 3 - Set fields value



Example 4 - Send user feedback with attachment



Example 5 - Send user feedback with attachment (async version)

The asynchronous nodes do the same as Example 4, but they do not need custom events. Each of them has its own **On Success** and **On Failure** execution pins. Thus the full graph is one chain.

1. **Create Feedback Async** - **On Success** gives the **Feedback** object.
2. **Set Field Value** - Set the values that the user typed in the form.
3. **Send Feedback Async** - Takes this **Feedback** object. On success, it returns the **Issue Id**.
4. **Send Attachment Async** - Takes the **Issue Id**, and the path, file name and mime type of the file to upload.

Example 6 - Get field values

